PRACTICAL - 2

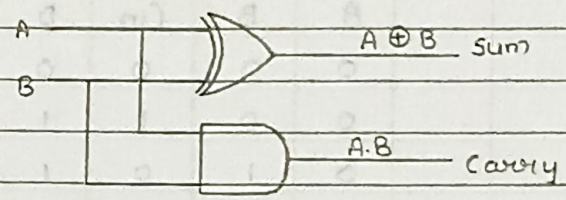
(A) Design and Implement of Half adder, Full adder and half Subtractor & full Subtractor.

(i) Half Adder

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$S = A'B + AB' = A \oplus B$

$C = AB$



(ii) Full Adder

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$S = A'B'C + A'BC' + AB'C' + ABC$

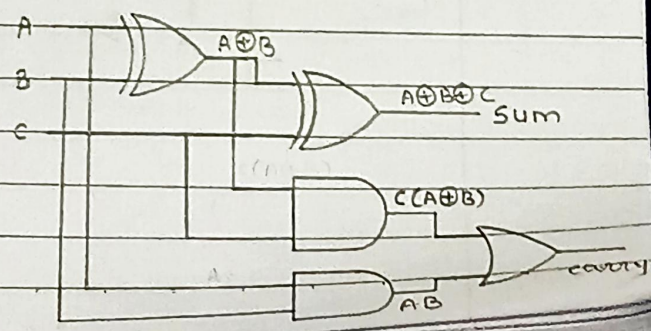
$= A'(B'C + BC') + A(B'C' + BC)$

$= A \oplus B \oplus C$

$Cout = \underbrace{A'BC + AB'C} + \underbrace{ABC' + ABC}$

$= C(A'B + AB') + AB(C + C')$

$= C(A \oplus B) + AB$

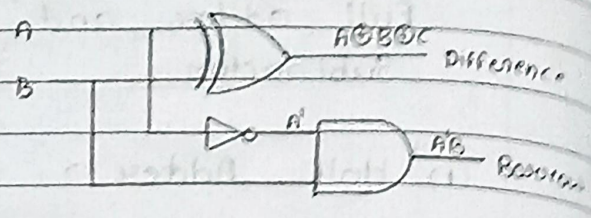


(iii) Half Subtractor

A	B	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$D = A'B + AB' = A \oplus B$

$B = A'B$



(iv) Full Subtractor

A	B	Cin	D	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$D = A'B'C + A'BC' + AB'C' + ABC$

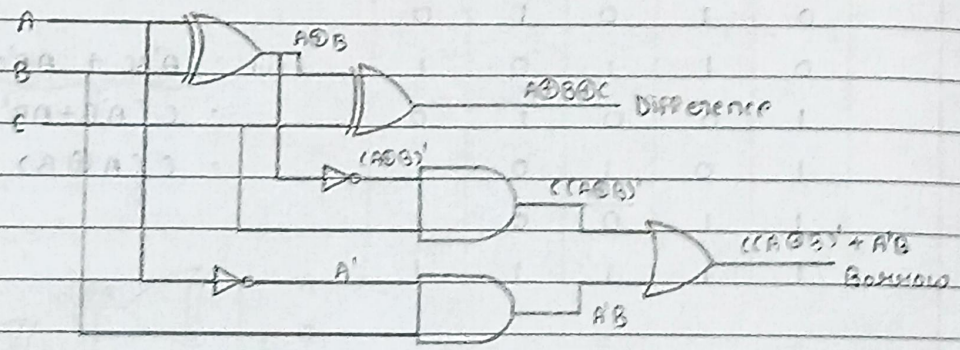
$= A'(B'C + BC') + A(B'C' + BC)$

$= A \oplus B \odot C$

$B = A'B'C + A'BC' + A'BC + ABC$

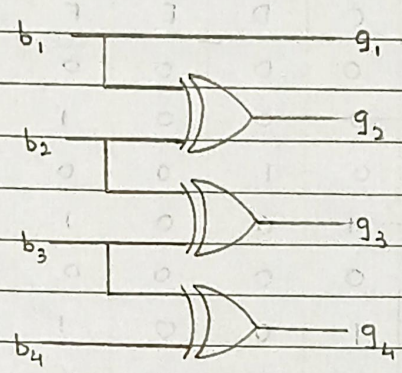
$= C(A'B' + AB) + A'B(C + C')$

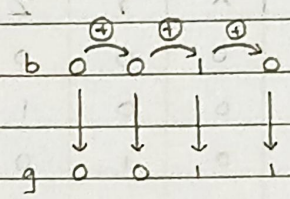
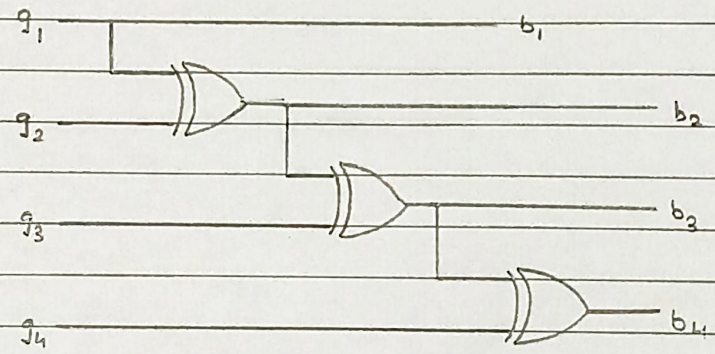
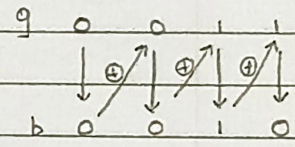
$= C(A \oplus B)' + A'B$



(A) Design the circuit to convert Binary to Gray code & vica versa.

(i) Binary to Gray Code





(ii) Gray Code to binary